Lab 2 – Gold Star Software Requirements Specifications

Version 1.0

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CS 411W: Professional Workplace Development II

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1. Introduction

1.1 Purpose

This is a document of detailed specifications and requirements for the Gold Star web-based application, designed to assist diagnosed persons with ADHD with completing everyday tasks.

1.2 Scope

Individuals with ADHD diagnoses or ADHD-like traits often struggle with consistent routine development. Some challenges include lack of focus, hyperactivity, and inattention. Gold Star is a web-based habit-building and time management platform tailored to those affected by ADHD, designed to help users organize their schedules, break down tasks, and build sustainable routines.

1.3 Definitions, Acronyms, Abbreviations

Activity Alternation - Switching from an activity requiring high mental focus to an activity requiring low mental focus.
Activity Switching - The cognitive process of disengaging from one task and shifting mental focus to another task.
Adaptive Scheduling -To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.
Application Programming Interface (API) - A set of standard rules, protocols, and definitions that allow different software applications to interact.
Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.
Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.
Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.
GitHub - A cloud-based platform where developers can collaborate on software code.
Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.
Hyperactivity - A state of being unusually or abnormally active.
HyperText markup Language (HTML) - The standard markup language used to create and structure webpages.
Javascript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.
Jest - An open-source Javascript testing framework.
JSDoc - An API documentation generator and syntax standard for JavaScript.
Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.
PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.
Reinforce - To make something stronger, more effective, or more compelling by adding support.
Relation Database Management System (RBMS) - software used to store, manage, query, and retrieve data structured in table form.
Rigid Scheduling - To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.
Routine Adaption - A requirement for an individual to change one's schedule or create new routines.
Routine - A sequence of actions regularly followed, or a fixed program.
Task Deviation - When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.
Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

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1.5 Overview

Section 2 delivers an overview of the Gold Star web application.
Section 2.1 gives an overall product perspective and purpose of the web application.
Section 2.2 describes key product features.
Section 2.3 defines user roles.
Section 3 will cover system requirements.

2. Gold Star Product Description

2.1 Product Perspective

Gold Star uniquely focuses on helping users organize tasks and schedules through an Adaptive Scheduling system which adjusts to missed tasks and changing priorities. It evolves with the user, reducing the frustration of falling behind, making it easier to stay on track. It utilizes techniques such as Habit Stacking, Activity Alternation, and the Pomodoro Technique which are recommended for people who struggle with ADHD-like symptoms. Users can choose their productivity level and alter their schedule at any time throughout the day, giving them enough flexibility to feel successful. Users can also group tasks by importance level, allowing Gold Star to become personalized to their unique scheduling needs.

2.2 Product Functions

Users can access the application through any off-the-shelf device with a modern web browser. The server side runs on a single hosting environment supporting the Node.js web server and PostgreSQL database, with deployments being automatically pushed through GitHub. The software to be developed includes the UI/UX built with HTML, CSS, and JavaScript. It will feature a calendar view, routine dashboard, productivity level system, and statistics dashboard. It will also feature a Node.js web server handling routing, logic, authentication components, and a WebSocket-based notification handler. A PostgreSQL database is included to store user account information, saved tasks, routine data, and sensitive user information kept in an encrypted form.

2.3 User Characteristics

Gold Star is an individually utilized based web application with the classification of a user, who will utilize Gold Star to help schedule and promote the success and completion of various tasks. Gold Star is designed for persons with ADHD but may be beneficial to persons of any background who struggle with staying on task or multitasking in general.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A